

Mark Joseph Valenzuela

AI Workflow Automation Engineer

n8n • AI Automation • AI-Assisted Development • API Integration • Infrastructure

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7+ years of professional experience, including 4+ years in IT across systems administration, infrastructure, and technical support — now specialising in AI workflow automation and AI-assisted software development. I build n8n workflows that wire LLMs, OCR, and messaging platforms into live business processes, ship production software by directing AI coding agents, and run the infrastructure underneath both.

Selected Projects

TRANSACTION MANAGEMENT MODULE — YAPII (AI REAL ESTATE SAAS)

TECHNOLOGIES: Claude Code, Node.js, Express, PostgreSQL, Drizzle ORM, React, Docker Swarm, Git Worktrees

- Rebuilt a brokerage's transaction management inside the company's AI SaaS platform to replace the paid third-party tool it depended on, matching it feature for feature.
- Directed six Claude Code sessions running in parallel, each isolated in its own git worktree and scratch database, coordinated into a single reviewable pull request.
- Built a trigger-based automation engine creating tasks from file conditions, plus intake automation that polls live web forms and applies the matching task template.
- Added 101 tests to code that previously could not be tested, and traced 236 suite failures to roughly five root causes.
- Enforced read-only access to every external system throughout, with tests that fail if write paths are re-enabled.

PLATFORM MIGRATION — RENDER TO SELF-HOSTED VPS WITH DOKPLOY

TECHNOLOGIES: Dokploy, Hostinger KVM 8, Docker Swarm, PostgreSQL, pg_dump, rclone, DNS/DMARC

- Provisioned a Hostinger KVM 8 VPS from scratch and deployed Dokploy as a self-hosted alternative to Render, preserving push-to-deploy for the team.
- Cut hosting from \$181 to \$25.64 per month — roughly \$1,860 a year — after evaluating and rejecting managed Postgres on cost.
- Cut over with zero downtime: deployed to a spare hostname, verified, then repointed the domain without restarting the container.
- Rehearsed the migration to a 51-second restore of a 7.2 GB dump across 416 tables before committing to it.
- Fixed a Dockerfile that had never successfully built, ended recurring hourly crashes by raising the Node heap limit, and replaced a paid backup add-on with scripted offsite dumps.

AI CUSTOMER SUPPORT CHATBOT

TECHNOLOGIES: n8n, LangChain AI Agent, OpenRouter, Facebook Graph API, Google Docs, Window Buffer Memory, Prompt Engineering

- Built an AI support chatbot for a pisonet business that answers customer enquiries from a Google Docs knowledge base without human intervention.
- Integrated Facebook Messenger with n8n and implemented automatic language detection to reply in English or Filipino.
- Engineered context-aware prompts so pricing and FAQ responses stayed accurate and consistent.

GCASH REPORTING AUTOMATIONS (BUY LOAD & SUMMER NANO)

TECHNOLOGIES: n8n, Telegram Bot API, OCR.space, Nominatim / OpenStreetMap, Google Sheets API, Google Drive, JavaScript

- Built end-to-end workflows automating GCash reseller reporting, with licence validation through Google Sheets before any request is processed.
- OCR'd the coordinates stamped on each installation photo, then checked they fall inside the Philippines before accepting the submission.
- Reverse-geocoded through OpenStreetMap and rejected entries where the detected location contradicted the city the installer typed.
- Recorded validated data into Google Sheets automatically and returned real-time confirmations, eliminating manual encoding.

Core Skills

AI & AUTOMATION

n8n • Workflow Automation • LangChain AI Agents & Chatbots • Prompt Engineering • OpenRouter • OCR.space • Nominatim / OpenStreetMap • Geofencing

INTEGRATION & DEPLOYMENT

REST APIs • Webhooks • JSON • Google Sheets API • Telegram Bot API • Facebook Messenger API • Dokploy • Docker & Docker Swarm • Linux VPS provisioning • DNS & DMARC

AI-ASSISTED DEVELOPMENT

Claude Code • Parallel agent sessions • Git worktree isolation • React • Node.js • Express • PostgreSQL • Python • PowerShell • JavaScript • Test-backed delivery

INFRASTRUCTURE & SUPPORT

Windows Server • Active Directory • Exchange Online • Microsoft 365 • VMware ESXi • Veeam Backup • TrueNAS • Synology • pfSense • Zabbix • Wazuh • Rundeck • iTop

Professional Experience

TECHNICAL VIRTUAL ASSISTANT, PRODUCT DEVELOPMENT — HOMEXA

August 2026 – September 2026

- Worked on Yapii, the company's AI-powered SaaS platform for real estate agents (Node.js, PostgreSQL, React).
- Delivered the transaction management module by directing parallel Claude Code sessions into a single reviewable pull request.
- Migrated the platform off Render onto a self-provisioned VPS running Dokploy, with zero downtime, and hardened destructive write paths to external systems.

WORKFLOW AUTOMATION ENGINEER — PRIVATE CLIENTS

2026 – Present

- Builds n8n automations integrating Telegram, Facebook Messenger, OCR, LLM agents, and Google Workspace for live businesses.
- Delivered the GCash reporting automations and an AI customer support chatbot running on a custom knowledge base.

IT INFRASTRUCTURE CONSULTANT — ST. CATHERINE DE NOVALICHES SCHOOL, INC.

2026 – Present • On-call

- Designed and deployed the school's network infrastructure: dual ISP with load balancing, structured cabling, switches, and wireless access points.
- Provides ongoing maintenance and technical support for administrators, faculty, and students.

IT CONSULTANT — MR. FREEZE ICE PLANT

2026 – Present • On-call

- Provides technical support, computer maintenance, Windows administration, and network troubleshooting for business operations.

SYSTEM ADMINISTRATOR — WOODFIELDS CONSULTANTS, INC.

June 2023 – February 2026

- **Servers & Network:** Provisioned and maintained physical and virtual servers, pfSense firewalls, network equipment, and NAS storage (TrueNAS, Synology).
- **Microsoft 365 & Active Directory:** Managed accounts, licensing, Exchange Online, access rights, and AD domain migrations (ADMT).
- **Backup & Monitoring:** Ran Veeam Backup & Replication, implemented Wazuh for threat hunting and Zabbix for infrastructure monitoring.
- **Support & Automation:** Handled hardware and software support (SAP, QuickBooks) via iTop, and automated repetitive administrative tasks using Python and Rundeck.

IT ASSISTANT — SUPER GROCERS INC.

October 2019 – October 2020

- Installed and configured hardware and software, diagnosed hardware, software, and network issues, and configured employee workstations.
- Maintained software licences, patched applications, and removed malware and ransomware.

Earlier: Field Sales Representative, Keymotors Inc. (2022–2023) • Marketing Assistant, PremiumBikes Corp. (2020–2022) • Accounting Staff, Sabrosa Foods Inc. (2019)

Education

BS Information Technology — STI College of Novaliches, June 2014 – May 2018